

Lying Latents: Auditing Distribution-Free Inference for Sparse Autoencoder Feature Discovery

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Advanced NLP, Monsoon 2026

1 Introduction

Language models are black boxes. To understand *why* a model behaves the way it does, researchers use sparse autoencoders (SAEs) to decompose internal representations into thousands of human-interpretable units called latents, each ideally corresponding to a single concept. The standard workflow then identifies which latent represents a concept of interest, intervenes on it, and declares it causally responsible if model behaviour changes accordingly.

This project asks whether that workflow actually supports the causal claims built on top of it. We argue it does not. Scanning a typical SAE dictionary at a standard significance threshold guarantees a large number of spurious discoveries by chance alone, and the field applies no correction for this. The most direct attempt to fix it (Enkhbayar, 2025) uses a statistical framework that requires knowing the distribution of SAE activations and makes an assumption that is structurally incompatible with how those activations actually behave. The formal guarantee therefore does not apply. Whether this breaks the procedure in practice is an open empirical question, and answering it is what this paper does.

Section 3 details the shortcomings that motivate this project; Section 5 states our specific contributions; and Sections 6 to 8 describe how we will produce them.

2 Background

Individual model components resist direct inspection: a single neuron typically responds to several unrelated concepts, a phenomenon attributed to *superposition* a model packs more concepts into its dimensions than it has room for, relying on their rare co-occurrence (Elhage et al., 2022). Sparse autoencoders (SAEs) undo this packing by compressing and reconstructing activations under a sparsity con-

straint, so that each *latent* (out of tens of thousands in a typical dictionary) ideally corresponds to one concept. Successive architectures (Bricken et al., 2023; Cunningham et al., 2023; Templeton et al., 2024) have mainly varied how sparsity is enforced, culminating in JumpReLU (Rajamanoharan et al., 2024). Most current work uses off-the-shelf public dictionaries such as Gemma Scope for Gemma 2 (Lieberum et al., 2024) rather than training its own.

Given a trained dictionary, the standard recipe for interpreting a latent is *search-then-validate*: *search* a labeled corpus for the latent whose activation best tracks a concept, then *validate* by forcing that latent on or off and checking that model behaviour shifts accordingly. Most “latent X is the concept- Y feature” claims are produced this way, and automated pipelines (e.g., Neuronpedia) run this recipe at scale. The main reason this matters is *steering*: clamping a latent believed to represent a concept can push generation toward it. The field’s evaluation infrastructure targets exactly these two steps (SAEBench for search, AxBench for steering), and only very recently has work begun importing statistical machinery from genetics and clinical trials to handle this kind of large-scale candidate screening the thread we continue.

3 Motivation

The search-then-validate workflow has two structural problems the field has acknowledged but not resolved with formal statistical machinery.

Multiple testing without correction.

Multiple testing without correction. A typical dictionary has 16,000+ latents. At a conventional 5% significance threshold, roughly 800 latents would appear to correlate with a random label even with zero true signal purely by chance. The standard pipeline applies no correction for this.

Data reuse between search and validation. The latent chosen for intervention is selected precisely because it scored highest on the search data, so re-measuring its effect on that same data is not an independent test: noise in its apparent correlation persists under re-measurement. Causal intervention does not rescue this ablating a chance-selected latent still perturbs the model, and with no “null intervention” reference distribution, an observed behavioural change carries little information about whether the original selection was meaningful.

Together, these mean claims from the standard pipeline carry no formal statistical guarantee, despite downstream uses (steering, safety auditing, factual editing) all presuming the latent is genuinely responsible for the concept rather than an artefact of an uncorrected search.

Enkhbayar (2025) applies the Model-X knockoff framework (Candès et al., 2018) to SAE latent selection, instantiating the required knockoff-generating distribution with a Gaussian surrogate. This is where our project begins. Under ReLU/JumpReLU activations, latents are non-negative and exactly zero for over 95% of inputs (Rajamanoharan et al., 2024). No Gaussian can represent an exact zero with positive probability, so the two distributions fundamentally disagree on which outcomes are even possible: their KL divergence is infinite. The robustness guarantee normally invoked to justify approximate distributional assumptions in the knockoff framework (Barber et al., 2020) bounds FDR inflation by exactly this KL divergence. The guarantee therefore does not apply here, because the bound itself does not exist. Whether this theoretical gap translates into actual FDR inflation in practice is an open empirical question, and answering it is the starting point of our project.

4 Related Work

The search-then-validate approach is a form of *selective inference*: a hypothesis is generated and tested on the same data, and the number of hidden hypotheses equals the size of the dictionary. Benjamini and Hochberg (1995) faced a similar issue in genomics and introduced false discovery rate (FDR) control as a practical way to handle large-scale testing instead of correcting for every hypothesis individually. The knockoff framework (Barber and Candès, 2015; Candès et al., 2018) builds on this by creating synthetic “knockoff” copies

of each feature that mimic its correlation structure. The knockoff framework (Barber and Candès, 2015; Candès et al., 2018) extends this by constructing synthetic *knockoff* copies of each feature that mimic its correlation structure while being provably independent of the response, giving a data-driven null with a finite-sample FDR guarantee; Barber et al. (2020) showed this guarantee degrades gracefully under a misspecified knockoff distribution, with FDR inflation bounded by the KL divergence between true and working conditional laws, provided it is finite. Enkhbayar (2025) is the first application of this framework to SAE latents, selecting 129 of 512 high-activity Pythia-70M latents on SST-2 at target FDR 0.1 via a Gaussian surrogate the direct prior for our work, whose zero-inflation mismatch we formalise and audit in Section 6.

Whether SAE-based steering reliably outperforms simple prompting baselines remains contested (Wu et al., 2025). A statistically grounded feature filter that predicts steering success would offer principled guidance on which latents are actually worth intervening on.

5 Contributions

Our project makes five contributions, in order of expected confidence, each directly addressing one shortcoming raised in Section 3.

(1) Empirical FDR calibration of the standard pipeline (Stage 1). Addressing the uncorrected multiple-testing problem, we quantify how often the conventional search-then-validate workflow certifies features under a global null, via label permutation. This is a direct, assumption-free error-rate measurement the field currently lacks, and yields a Westfall–Young style multiplicity correction usable independently of the knockoff framework.

(2) Falsifiability audit of Model-X exchangeability (Stage 2). Addressing the unverified Gaussian assumption underlying Enkhbayar (2025), we test whether the exchangeability property required for valid knockoffs holds for real SAE latents, via zero-mass, two-sample, and higher-moment diagnostics. This tells the field concretely where and how severely the Gaussian assumption breaks, independent of its downstream consequences.

(3) Realised FDR on semi-synthetic ground truth (Stage 3). Addressing the fact that the theoretical failure of the KL bound does not by itself

establish practical harm, we use real activations with planted synthetic signal to measure whether the Gaussian knockoff procedure’s realised FDR matches its nominal target. Either outcome (robustness despite misspecification, or genuine inflation) is a useful, reportable finding.

(4) Distribution-aware knockoff procedures (Stage 4). Addressing the need for a working alternative if the Gaussian surrogate does fail, we propose and benchmark three repairs (a hurdle knockoff, a binarised variant, and an e-value sample-splitting fallback) on the Stage 3 benchmark, establishing whether distribution-aware construction improves FDR control in practice.

(5) Predictive validity of filter survival for steering (Stage 5). As a downstream check connecting the statistical question to its practical motivation, we test whether filter-surviving features are more predictive of successful activation steering than uncorrected features.

6 Methodology

Setup. Let $X \in \mathbb{R}^{n \times p}$ denote SAE latent activations over n inputs, and let Y denote the task label. Following [Enkhbayar \(2025\)](#), we retain high-activity latents by firing rate ($p \approx 512$ to 2048). A latent j is null if $Y \perp X_j \mid X_{-j}$ (conditionally, not marginally). This is the gap the standard pipeline ignores: search tests marginal association, while the claim made is conditional, so a latent that merely co-fires with the true carrier passes screening. We fix all thresholds and comparison sets before running each stage.

Stage 1: Calibrating the standard pipeline. We permute task labels $B \approx 500$ times, destroying all association while preserving the latent covariance exactly, and run the conventional workflow end to end on each permutation: score every latent (mean activation difference, AUROC, probe weight), take the top- k , ablate, and record whether the behavioural change clears a pre-registered threshold. Since every latent is null under permutation, the relevant quantity is the family-wise error rate under the global null, not the FDR, which degenerates to 1 whenever anything is rejected. The permutation null distribution of the maximum latent score is exactly a Westfall–Young max-statistic correction.

Stage 2: Auditing the Model-X assumption.

The knockoff guarantee is distribution-free only with respect to $Y \mid X$ ([Candès et al., 2018](#)); it still demands exact knowledge of the law of X . Assuming Gaussianity yields second-order knockoffs: only the first two moments are matched. We test three necessary consequences of exchangeability on real Gemma Scope latents: (i) the zero-mass discrepancy, where $\Pr(X_j = 0) > 0.95$ under JumpReLU versus 0 under any Gaussian, so the indicator $1[x = 0]$ separates real from knockoff almost perfectly; (ii) classifier and MMD two-sample tests on swapped versus unswapped copies, reporting AUC against the exchangeable baseline of 0.5; and (iii) skewness, kurtosis, and tail quantiles beyond what second-order matching attempts.

Stage 3: Realised FDR against known ground truth.

Realised FDR is uncomputable on real data because its numerator requires knowing which discoveries are false. We keep the real activation matrix X (preserving zero-inflation, tails, and correlation structure) and generate synthetic labels $Y = f(X_S) + \varepsilon$ from a random subset S , $|S| \in \{10, 20, 30\}$, using linear-logistic and interaction-containing f across a range of signal-to-noise ratios. Latents outside S are conditionally null by construction. We run Gaussian knockoff+ at $q \in \{0.05, 0.1, 0.2\}$ and record realised FDR and power.

Stage 4: Distribution-aware knockoff procedures.

Hurdle knockoffs (primary) model each latent as a two-component conditional, logistic for $\Pr(X_j > 0 \mid X_{-j})$, log-normal or Gamma for the positive part, sampled via sequential conditional independent pairs ([Candès et al., 2018](#)). Once the working distribution has an atom at zero, absolute continuity is restored, the KL divergence becomes finite, and the [Barber et al. \(2020\)](#) bound can be numerically estimated for the first time in this setting. *Binarised knockoffs* run on $Z_j = 1[X_j > 0]$, avoiding the atom problem by construction at the cost of power. *Sample splitting with e-values* ([Wang and Ramdas, 2022](#)) discovers on one split and tests on another, remaining valid under adaptive selection with no distributional assumption, serving as a guaranteed fallback. All three are evaluated on the Stage 3 benchmark against the Gaussian baseline.

Stage 5: Does survival predict steering?

Filter survivors are compared against matched non-survivors and random latents, matched on firing

frequency and decoder norm. We measure target-token probability shift on templated prompts as the primary effect, and WikiText perplexity plus clamped/unclamped KL divergence as collateral, sweeping the clamp coefficient as a multiple of each latent’s observed maximum.

Scope and compute. Gemma-2-2B with Gemma Scope (Lieberum et al., 2024) is the primary setting; Pythia-70M replicates Enkhbayar’s exact configuration for direct comparability. Activations are cached once; all analysis beyond Stage 5 is CPU-bound, and Stage 5 requires only forward passes. No model training is involved. Priority order: Stages 1 to 2 are the minimum viable result, Stage 3 with the e-value fallback is the core contribution, and the hurdle construction and steering evaluation are stretch goals in that order.

7 Dataset

Labeled classification tasks. We retain SST-2 (Socher et al., 2013) to directly reproduce and critique Enkhbayar (2025). Since a single binary concept yields only one search-validate trial, our primary source is the SAEbench sparse probing suite: 35 binary tasks across five datasets (Bias in Bios, Amazon Reviews, AG News, EuroParl language ID, GitHub language classification), each with a fixed 4,000/1,000 train/test split truncated to 128 tokens. This gives 35 independent trials and keeps our results comparable to the canonical SAE evaluation setup, with a built-in difficulty gradient: EuroParl/GitHub concepts serve as power positive-controls, while Bias in Bios and Amazon categories are the regime where false discoveries should actually appear.

Unlabeled activations, synthetic ground truth, and steering data. For the knockoff audit (Stage 2) we use the Pile (Gao et al., 2020) for Pythia and FineWeb for Gemma-2-2B, restricting knockoff generation to the top few hundred latents by firing rate given the intractability of estimating a full 16,000-dimensional conditional law, and reporting the n/p ratio explicitly. For Stage 3 ground truth, we derive both a permutation null and a planted-signal benchmark directly from the labeled tasks above. For the downstream steering check (Stage 5), we use AxBench (Wu et al., 2025), built on Gemma-2-2B and Gemma Scope labels, and verify in advance that its layer and SAE width match our audit configuration.

8 Evaluation Plan

We evaluate each of our five methodological stages against specific quantitative criteria to determine whether current practices fail and whether our proposed repairs succeed.

Stage 1: Pipeline Calibration. We measure the Family-Wise Error Rate (FWER) across $B = 500$ label permutations. The primary output is the empirical null curve of maximum latent scores. Success here is the successful establishment of a direct, assumption-free estimate of the standard pipeline’s false-certification rate.

Stage 2: Model-X Exchangeability Audit. We evaluate the Gaussian knockoff assumption using two-sample tests. The metrics are the swap-classifier Area Under the Curve (AUC) and the empirical zero-mass probability gap ($\hat{p}_0$). An AUC significantly greater than 0.5 or a substantial zero-mass gap robustly falsifies the exchangeability assumption.

Stage 3: Ground Truth Realised FDR. Using semi-synthetic planted signals, we compare the realised False Discovery Rate (FDR) and statistical power against the nominal target $q \in \{0.05, 0.1, 0.2\}$. If the realised FDR tracks the nominal q , the Gaussian surrogate is robust despite its theoretical flaws; if it significantly exceeds q , empirical repair is necessary.

Stage 4: Distribution-Aware Knockoffs. We evaluate our proposed hurdle, binarised, and e-value fallback methods using the same FDR and power metrics as Stage 3. A successful repair will demonstrate tighter adherence to the nominal FDR target or strictly higher statistical power compared to the Gaussian baseline.

Stage 5: Downstream Steering Validity. We assess practical utility by measuring target-token probability shift (primary) alongside WikiText perplexity and KL divergence (collateral damage). Success is defined as filter-surviving latents producing a statistically significant improvement in steering efficacy (one-sided Wilcoxon signed-rank test) compared to frequency-matched non-survivors.

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